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Dissertation Title

"Leveraging Text Analytics for Customer Insight: A Study of Market Strategy Improvement through Online Reviews and Social Media Feedback – A Case Study of Scottish Power"

Declaration

I certify that this dissertation is my own work. I have read the University regulations concerning plagiarism.

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Abstract:

This research explores the impact of online reviews and social media feedback on consumer behavior, emphasizing the integration of text analytics to improve customer satisfaction and business decision-making. As online reviews increasingly influence purchase decisions, businesses can leverage the abundant unstructured data to gain actionable insights. The study identifies a gap in the systematic use of customer feedback, highlighting the potential of text analytics techniques like sentiment analysis and thematic analysis to enhance service delivery and marketing strategies.

Using a qualitative mono-method approach, the study analyzes archival data from Trustpilot and employs tools like NVivo and Voyant for comprehensive analysis. Positive reviews often focus on excellent customer service, smooth operations, and affordability, while negative reviews revolve around poor customer service, billing errors, and unresolved complaints. Sentiment analysis shows that although most feedback is positive (64%), negative reviews (28%) significantly affect customer loyalty and profitability.

To address these challenges, the research proposes a framework combining advanced analytics with structured feedback mechanisms. Recommendations include adopting predictive analytics for proactive service improvements, enhancing pricing transparency, and prioritizing customer-centric training programs. The European Foundation for Quality Management (EFQM) model is suggested as a strategic tool for aligning operations with customer expectations, promoting continuous improvement, and boosting organizational performance.

The findings emphasize the critical role of customer feedback in driving business innovation and the need for data-driven strategies to maintain customer loyalty and competitiveness. By utilizing text analytics and structured frameworks, businesses can enhance customer experiences, optimize services, and achieve long-term profitability. This research contributes to understanding customer feedback utilization and provides actionable insights for organizations operating in dynamic consumer markets.

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1. Introduction:

1.1 Research background:

Online reviews and social media channels have emerged as key determinants in influencing consumer behavior. Multiple studies reveal that 93% of buyers refer to online reviews for purchase decisions, placing more confidence in them than in customary marketing techniques (Ahn & Lee, 2024). Reviews are a growing source of information for consumers about products. Online shopping websites include reviews from previous customers who have already bought the product, which are compared to the vendors' official product information (Baek et al., 2012). Additionally, there is a growing tendency of customers sharing their buying experiences on social media (Floh et al., 2013).

According to research, 93% of consumers believe that internet reviews will influence their purchasing decisions. This suggests that the majority of consumers consistently read online reviews and base their judgments on the comments (Vimaladevi and Dhanabhakaym, 2012). A consumer's decision to buy after reading internet reviews is the result of a psychological process that combines information processing and vision. It is clear from the literature that most studies have concentrated on the results and effects of online reviews on decisions to buy, but they have not given as much insight into the fundamental mechanisms that shape consumer perception (Sen and Lerman, 2007).

Social media and internet reviews have made it essential for businesses to get feedback from customers in order to better understand their impressions and make better decisions (Maryam Abdullah AL-Barrak & A. Al-Alawi, 2024). Sentiment analysis and text analytics are efficient ways to glean insightful information from unstructured customer feedback data (Amani K. Samha et al., 2014). These methods can be used to find sentiment patterns, user opinions, and product features, allowing companies to improve their goods and services (Atika Qazi et al., 2013).

A suggested framework for review analysis seeks to categorize reviews into several kinds, giving consumers and product creators pertinent information (Atika Qazi et al., 2013). Furthermore, in particular businesses, like the hospitality sector, latent semantic analysis (LSA) can be utilized to pinpoint the essential characteristics influencing client happiness and discontent (Xun Xu et al., 2017).

1. Customer Influence and Decision-Making:

Online reviews and social media input have a significant impact on customer decisions, making these platforms important players in determining consumer behavior. A brand's reputation can be greatly impacted by positive or unfavorable reviews, which can either draw in new business or turn off existing clients.

To preserve their reputation, companies must actively track and address this comments. Participating in online customer reviews contributes to the upkeep of a good reputation and offers insightful information for enhancing offerings and client happiness.

2. Underutilization of Customer Insights:

The potential of client feedback via internet platforms to promote service improvements is not being properly utilized by many firms. Despite the wealth of data available through reviews and social media, much of it stays unexplored, leaving businesses without the useful insights they could be gaining.

It is necessary to close the gap between the availability of feedback and the production of useful insights. Companies need to use more efficient techniques to evaluate this data and turn it into strategies that improve consumer happiness and services.

3. Text Analytics as a Solution:

Text analytics provides a productive way to handle unstructured data from social media and online reviews. It can extract insightful information from vast amounts of client feedback—too much to study manually—by utilizing sophisticated algorithms.

This strategy aids companies in identifying areas of discontent, developing trends, and client pain spots. With these insights, businesses can make data-driven decisions to enhance their offerings, quickly resolve problems, and exceed client expectations.

4. Research Problem:

There aren't many organized frameworks for using text analytics to examine client feedback from social media and internet reviews in an efficient manner. While text analytics tools are accessible, many firms do not have organized ways to properly exploit these technology for assessing customer sentiment and feedback.

Because of this, businesses frequently find it difficult to turn text analytics results into usable service enhancements. Businesses lose out on opportunities to improve their services and effectively address the wants and concerns of their customers by not having clear frameworks in place.

5. Business Need for Service Improvement:

In today's fast-paced marketplaces, firms must constantly upgrade their services to be competitive. Businesses constantly need to improve their services to keep up with the continuously evolving expectations of their customers. Businesses run the danger of losing their competitive edge and slipping behind without constant development initiatives.

A potent strategy for promoting these advancements is to make use of client feedback information. Businesses can find areas for improvement and modify their services to better suit client wants by examining input from social media and online reviews. This improves customer satisfaction and experience.

6. Bridging the Gap:

The goal of this research is to create a framework for the efficient integration of text analytics insights into business decision-making procedures. Businesses can methodically utilize the abundance of information found in client feedback via social media and online reviews by implementing an organized strategy.

The framework's main goal is to assist businesses in more effectively interpreting and responding to this digital input. Businesses may close this gap by transforming insights into practical initiatives that improve services, cater to consumer wants, and eventually boost overall performance and customer happiness.

1.2. Research Aim:

The dissertation seeks to leverage text analytics to gain insights from customer feedback via online reviews and social media. This involves analyzing large volumes of unstructured text data to enhance marketing strategy quality and customer satisfaction.

1.3. Research Objectives:

1. To implement text analytics-driven marketing adjustments:

Apply marketing adjustments driven by text analytics with the goal of improving Scottish Power's online and social media audiences' perceptions. Scottish Power can discern important themes and recurrent issues that impact public discussions about the company by scrutinizing customer feedback obtained from online reviews and social media posts. The knowledge acquired will guide focused marketing tactics, enabling the business to solve client issues, highlight its advantages, and improve client interaction.

2. To improve Scottish Power's market strategy effectiveness:

Enhance Scottish Power's market plan efficacy by fine-tuning service offerings with data-driven insights. Customer feedback analysis can be used to pinpoint preferences and problems, enabling customized changes that improve audience interactions and increase engagement on social media platforms. The goal of this customer-centric strategy is to develop a more adaptable marketing plan.

3. To enhance Scottish Power's customer satisfaction:

Improve customer satisfaction at Scottish Power by identifying and resolving recurring issues found in customer feedback. By analyzing online reviews and social media posts, the company can uncover frequent concerns and areas for enhancement. Directly addressing these issues through focused service improvements will result in better customer experiences and increased satisfaction. This strategy is designed to build stronger customer relationships and promote long-term loyalty.

1.4. Research Questions:

1. How customer opinion on business metrics such as loyalty and profitability can impact an organisation's marketing strategy?

Text mining tools like Voyant and NVivo are useful for analyzing large amounts of unstructured customer feedback by identifying important themes and sentiments. Through sentiment analysis, feedback is categorized as positive, negative, or neutral, providing insights that can influence an organization's marketing strategy. Positive feedback suggests high customer loyalty and profitability, guiding marketing to focus on retention and advocacy. On the other hand, negative feedback pinpoints areas that need improvement, helping businesses refine their marketing strategies to address customer issues, reduce churn, and improve profitability by enhancing the customer experience.

2. What recurring patterns and insights can be identified through thematic analysis of customer feedback, and how do they influence customer satisfaction and business performance?

Thematic analysis of customer feedback uncovers recurring patterns, such as concerns with service quality, product performance, and the responsiveness of customer support. These insights are vital for gauging customer satisfaction, as they reveal both strengths and areas requiring attention. By tackling these common issues, businesses can improve the customer experience, increase retention, and ultimately enhance business performance by aligning their services more effectively with customer needs and expectations.

3. How can data-driven customer insights optimize service delivery and target areas for improvement?

Leveraging data-driven insights from customer feedback, visualized using tools that showcase the relationships among frequently mentioned terms, helps businesses pinpoint critical issues in service delivery. By recognizing the links between customer concerns and recurring themes, organizations can concentrate on particular areas that need enhancement. These insights facilitate the creation of targeted strategies aimed at improving service delivery, enriching customer experiences, and boosting overall satisfaction.

4. How can organizations leverage insights derived from text analytics to enhance decision-making processes and optimize the customer experience by addressing feedback?

Integrating insights from text analytics into a data-driven business model can influence strategic decisions concerning service improvements, resource allocation, and customer engagement. For example, a persistent trend of negative feedback regarding product delivery may result in increased investments in logistics. Actionable recommendations could include providing staff training based on customer service input, optimizing product selections, and enhancing communication strategies to improve the overall customer experience.

1.5. Research Structure:

The dissertation is structured into five components: the Introduction, which contextualizes the study by describing its background, knowledge deficit, inquiries, goals, and objectives; the Literature Review, reviewing appropriate academic sources, leading researchers, and Text Analytics application tools; the Methodology, elucidating the research design, philosophical underpinning, data collection, and analysis procedures; the Analysis and Findings, divulging valuable insights and tackling research questions; and the Conclusion, summing up vital findings, providing recommendations, discussing study limitations, and proposing areas for further exploration.

2. Literature Review:

Text mining is the process of analyzing and processing unstructured and semi-structured material using technology. In order for algorithms to be used to assess the data, it is an effort to transform the textual data into a structured and quantitative format (Miner et al. 2012). Information retrieval, document classification, online link analysis, and natural language processing are all applications of text analytics (Thakur and Kumar, 2022).

Initially developed for information retrieval and summarization, with origins dating back to Thomas Hyde’s 1674 library catalog for Oxford's Bodleian Library, text mining now focuses on extracting information to uncover hidden patterns in text data (Lamba & Madhusudhan, 2022). Text analytics-based theory-building approaches are becoming more and more popular as management scholars use text mining and big data analytics to extract insights and develop theories from the abundance of unstructured textual data from online platforms like news stories, scholarly publications, and social media (Kar & Dwivedi, 2020).

The increasing number of users of the massive data resources maintained on digital platforms that support these services underscores the significance and viability of text mining in the field of service management (Kumar et al., 2021). Numerous industries, including IT, healthcare, political document evaluation, energy, and services, employ text mining extensively to quickly identify important insights and extract useful information from massive amounts of pre-processed unstructured data (Chu et al., 2020).

2.1. Historical Roots of Text Analytics in Marketing:

In marketing, early Natural Language Processing (NLP) applications were constrained by limited computing power, restricting its potential for analyzing unstructured customer feedback, but advancements in digital data collection, storage, and computing technology since the early 2000s have enabled marketers to efficiently analyze extensive online content, understand customer preferences, and create more targeted strategies (Duan et al., 2020).

Since its inception in the 1950s, when Natural Language Processing (NLP) and Artificial Intelligence (AI) made significant strides (Moreno-Sandoval & Redondo, 2016), text analytics in marketing has developed into a vital tool for extracting insights from a variety of textual data sources. Businesses utilize these sources, which include social media posts, customer reviews, and other unstructured data types, to better understand consumer behavior, enhance customer experiences, and hone marketing tactics (Berger et al., 2019). A variety of advanced methods are used in this field, such as named entity recognition to categorize important entities like names or locations, information extraction to find pertinent data, and machine learning to improve pattern recognition and predictive analytics (Moreno-Sandoval & Redondo, 2016).

The advancement of marketing analytics is attributed to the collaboration of disciplines such as marketing, expert systems, and statistical analysis, unifying an array of methodologies to effectively handle intricate business issues (France & Ghose, 2018). Recent developments have addressed big data challenges by processing large-scale unstructured datasets and utilizing cutting-edge computational techniques to improve accuracy and decision-making in dynamic markets. Marketing analytics, which offers strong tools for data visualization, customer segmentation, and predictive modeling, helps businesses to derive actionable insights (France & Ghose, 2018; Wedel et al., 2016).

2.2. Development of NLP and Sentiment Analysis:

Sentiment analysis and natural language processing (NLP), which have gained popularity recently, are used in a variety of contexts, including improving voice assistants, examining social media content, and figuring out customer sentiments to enhance user experiences and decision-making (Sharma, 2021). In addition to giving researchers useful information for comprehending societal behaviors and attitudes, NLP approaches are being used more and more to evaluate public sentiment on platforms such as Twitter. This allows businesses to watch trends, measure consumer perceptions, and improve their tactics (Hasan et al., 2019).

For sentiment analysis, a variety of natural language processing (NLP) techniques can be used, including opinion mining to extract subjective information, text classification to group data into categories, and lexical analysis to look at word patterns and context. The techniques chosen will depend on the particulars of the data and its level of complexity (Panchal et al., 2022). By identifying key textual patterns and contextual relevance, techniques such as Bag of Words (BoW), which depicts text as a collection of word occurrences, and Term Frequency-Inverse Document Frequency (TF-IDF), which emphasizes the significance of particular terms within a document in relation to a dataset, have demonstrated efficacy in improving sentiment classification accuracy (Hasan et al., 2019).

2.3. Influence of User-Generated Content (UGC) and Online Reviews:

User-generated content (UGC) significantly influences consumer conduct in digital settings by delivering insights on alternative products, accentuating choices with greater perceived worth or utility, and acting as a marker for product excellence, which consequently impacts buying decisions and brand outlook (Wang et al., 2013).

Online reviews, notably those from influencers, considerably impact purchase intentions due to their perceived effort and trustworthiness, making them more consequential than other review types (Puspitasari & Aruan, 2023). In the tourism sector, reviews significantly affect accommodation decisions, with research demonstrating that female consumers are particularly swayed by these reviews, notably concerning enjoyment and travel planning inspiration (Gretzel & Yoo, 2008). Gender discrepancies are also apparent in the impact of user-generated content (UGC) on purchase intentions and trust, with females generally being more receptive to such content (Sethna et al., 2017). Nonetheless, gender disparities are not discernible in behaviors like inclination to read UGC, purchase frequency, or penning reviews for products deemed gender-neutral (Sethna et al., 2017).

2.4. Core Techniques in Text Mining for Customer Feedback:

Text mining has evolved into a vital technique for scrutinizing customer feedback and augmenting product quality by extracting valuable insights from massive volumes of unstructured textual data. Leveraging a blend of statistical approaches, machine learning, and natural language processing (NLP) methods, text mining facilitates businesses in uncovering latent patterns and trends that would be arduous to ascertain manually (Rangu et al., 2017). This strategy empowers organizations to attain an exhaustive understanding of customer experiences, viewpoints, and inclinations at a broad scale, enabling rapid assessment of customer emotions, identification of product flaws, and monitoring of nascent trends (Gamon et al., 2005).

Despite the intricacies of evaluating customer feedback due to the data's diversity and magnitude, text mining furnishes an effective methodology for processing and scrutinizing this information (Ordenes et al., 2014). Opinion mining, a critical branch of text mining, encompasses tasks like sentiment categorization, where feedback is grouped according to positive or negative emotions, theme extraction, which recognizes central topics discussed in feedback, and data visualization, which presents findings in a readily comprehensible arrangement (Lee et al., 2008). Furthermore, sophisticated frameworks such as the ARC (activities, resources, and context) model provide a more comprehensive perspective of customer experiences by contemplating the broader framework in which feedback occurs (Ordenes et al., 2014). By capitalizing on these methods, enterprises can not only augment product caliber but also bolster customer satisfaction by tackling concerns and adapting to consumer requirements more adeptly (Rangu et al., 2017; Gamon et al., 2005).

2.5. Role of Machine Learning in Enhancing Text Analytics:

Machine learning (ML) has considerably revolutionized text analytics, offering potent tools for the examination and processing of substantial volumes of unstructured textual data with outstanding precision and efficiency. These ML techniques augment various facets of text analysis, encompassing text classification, which involves grouping documents into predefined classes, sentiment analysis to measure emotional tone, and predictive modeling for forecasting patterns or outcomes based on textual data (Sangwan & Bhatnagar, 2020). The versatility of ML has facilitated its implementation across numerous domains. For example, ML-driven text analytics assists in optimizing operations and boosting justice dispensation in the legal sector by automating tasks like case sorting and legal inquiry (Saha et al., 2020). Likewise, in academic environments, it has been leveraged to assess faculty efficacy by scrutinizing student comments and institutional reports (Pramod et al., 2022).

The incorporation of ML into text analytics has also demonstrated its worth in scrutinizing social media data, an abundant source of unstructured text. This aptitude fosters better decision-making across domains such as customer feedback examination, where enterprises can detect consumer inclinations and dissatisfaction, and risk management, where looming threats or issues can be discerned promptly (Garg & Sharma, 2020). Moreover, scholars have advanced comprehensive classification frameworks to simplify the utilization of ML in text analytics, furnishing structured guidance for forthcoming investigations and deployments in this domain (Sangwan & Bhatnagar, 2020).

Despite its transformative potential, text analytics using ML faces challenges such as complex data preprocessing and managing large datasets, necessitating advanced algorithms and computational techniques to extract meaningful insights and meet evolving analytical demands (Garg & Sharma, 2020).

2.6. Impact of Text Analytics on Customer Insight

Text analytics and natural language processing (NLP) play a critical role in extracting valuable insights from unstructured textual data (Yusupha Sinjanka, 2023). These tools help businesses analyze diverse data sources, such as chat logs and social media posts, to understand customer sentiments and predict their behavior and preferences (S. Garg & S. Singh, 2018). By applying text mining and sentiment analysis on large document collections, businesses can gain customer intelligence, optimize operations, and identify improvement opportunities for products or services (G. Chakraborty, 2014).

Text analytics provides valuable insights into consumer behavior and emerging trends, helping businesses maintain a competitive edge (Berger et al., 2019). Methods such as term document frequency matrices and association rule mining further improve capabilities, enabling automated prediction of customer feedback in real-time applications like chat logs (S. Garg & S. Singh, 2018). These scalable analysis techniques facilitate rapid response to customer needs, ultimately enhancing overall customer experiences and satisfaction.

Although text analytics holds great promise, it also presents challenges, particularly concerning internal and external validity, which require attention to ensure accurate and useful interpretations of results (Berger et al., 2019). As researchers and practitioners improve these methods, businesses can leverage text analytics to inform decisions, optimize marketing strategies, and maintain competitiveness in a rapidly changing market (Yusupha Sinjanka, 2023).

2.7. Social Media Analytics for Real-Time Insights:

Social media analytics has become essential for businesses and researchers to extract valuable insights from unstructured data generated on platforms such as Twitter and Facebook (Abrol et al., 2015; Kumar & Sinha, 2016). By examining real-time social media streams, organizations can discover patterns, trends, and user sentiments reflecting public opinion and market dynamics, applicable to security, marketing, and societal issues (Abrol et al., 2015).

To efficiently handle social media data complexities, researchers have developed advanced frameworks integrating parallel processing, caching mechanisms, and messaging queues for scalable data workflows (Singh & Verma, 2020). These frameworks enable real-time analysis of large datasets.

Machine learning algorithms are crucial in social media analytics, enhancing sentiment analysis, trend prediction, and classification tasks. Algorithms like SVM and LSTM networks offer efficient solutions for processing sequential data, allowing businesses to make data-driven decisions and improve customer engagement (Singh & Verma, 2020).

Ensuring accurate and relevant insights requires attention to data quality and analytical focus, including removing irrelevant content and defining clear analysis objectives (Ganis & Kohirkar, 2015). When executed well, social media analytics empowers organizations to harness the potential of digital platforms, driving innovation, improving decision-making processes, and maximizing value from dynamic data sources (Ganis & Kohirkar, 2015).

2.8. Aspect-Based Sentiment Analysis for Granular Insights

Aspect-Based Sentiment Analysis (ABSA) is an advanced technique that provides a detailed understanding of customer opinions by identifying and evaluating sentiments related to specific attributes or aspects of products, services, or experiences. Unlike traditional sentiment analysis that broadly classifies sentiments as positive, negative, or neutral, ABSA enables businesses to pinpoint customer preferences and concerns at a granular level (Pavlopoulos & Androutsopoulos, 2014).

One notable development in ABSA research is multi-granularity aspect aggregation, which addresses the challenge of grouping similar aspects into coherent categories for analysis. By aggregating aspects at different granularities, businesses can identify both broad trends and detailed insights, resulting in more accurate sentiment evaluation (Pavlopoulos & Androutsopoulos, 2014).

Another ABSA innovation is the integration of structural learning models, which combine sentiment classification and aspect identification into a single framework. This approach enables simultaneous analysis of sentiments and aspects at various levels of detail, allowing for a more comprehensive understanding of customer feedback (Fang & Huang, 2012).

For non-English languages like Chinese, researchers have developed specialized ABSA models that operate at different linguistic layers, such as radical, character, and word levels. These language-specific adaptations have outperformed existing ABSA models, highlighting the importance of customizing approaches for diverse linguistic and cultural contexts (Peng et al., 2018).

In conclusion, ABSA offers businesses and researchers a powerful tool for extracting granular insights from customer sentiment data, which can inform targeted decision-making and strategies. As ABSA continues to advance with innovations like multi-granularity aspect aggregation, structural learning models, and language-specific adaptations, its potential to provide nuanced, actionable insights will grow, making it an invaluable tool for analyzing customer feedback across various industries and languages.

2.9. Integration with CRM:

Effective Customer Relationship Management (CRM) integration is vital for improving corporate performance and ensuring CRM project success (Meyer & Kolbe, 2005). Integrating CRM systems streamlines processes, centralizes customer information, and enhances customer interactions, driving competitive advantage. However, a lack of consistent definition and theoretical framework in CRM integration research contributes to fragmented understanding (Meyer, 2005), posing challenges for successful CRM implementation.

Several studies have explored integrating CRM with other business systems like collaboration tools and enterprise resource planning (ERP) systems to enhance customer data sharing and cross-departmental coordination (Surrey, 2012). Seamless integration is key to maximizing CRM potential, with collaboration systems fostering better communication and decision-making, ultimately improving customer service.

To bridge research gaps and improve CRM practices, a multidisciplinary approach is necessary. This includes blending theories from information systems, organizational behavior, and marketing, and considering integration at both project and enterprise levels to align CRM strategies with broader business objectives (Meyer & Kolbe, 2005; Meyer, 2005). This comprehensive approach provides organizations with a robust foundation for overcoming CRM challenges and boosting performance.

2.10. Predictive Analytics and Trend Forecasting Using Text Data:

Organizations looking to foresee future changes and make data-driven decisions now depend heavily on predictive analytics and trend forecasting using text data. Researchers have created a number of techniques to spot new trends and forecast how they will grow over time, usually by fusing visual analytics, machine learning, and text mining. Large amounts of unstructured data can be mined for useful information using text mining techniques, and machine learning models can be used to find patterns and links in the data to predict trends. By enabling users to examine and interpret data in intuitive ways, visual analytics—which combines interactive visualizations with statistical and machine learning techniques—has further improved trend detection (Secco et al., 2023).

One popular trend forecasting approach is association analysis, which helps find connections between different topics, terms, or events likely to evolve in the future. For instance, association analysis has been used in data mining and machine learning to discover emerging research topics and predict their potential impact (Hurtado et al., 2016). Ensemble forecasting, which merges the outputs of multiple models, is another technique used to enhance prediction accuracy, particularly for complex and dynamic datasets.

These predictive analytics techniques have been applied across industries like finance, healthcare, and marketing. In finance, they can forecast market movements, while in healthcare, trend forecasting can predict the spread of diseases or new health concerns. In marketing, forecasting helps businesses stay ahead of consumer preferences and adjust strategies accordingly (Agarwal & Tiwari, 2024).

Predictive analytics empowers organizations to respond proactively and maintain a competitive edge in a data-driven world. The ability to extract insights from large datasets, predict future developments, and adapt is crucial for businesses and policymakers navigating complex and rapidly changing environments. As predictive analytics continues to evolve, incorporating advanced methods like deep learning and real-time data processing promises more accurate and timely trend forecasts in the future.

2.11. Opinion Mining and Summarization:

Opinion mining and summarization techniques are crucial for analyzing and condensing user-generated content from various online sources, providing valuable insights into consumer preferences and sentiments. These methods help businesses understand customer satisfaction, identify improvement areas, and inform decision-making processes (Vijay B. Raut, 2014).

Opinion classification techniques include machine learning and lexicon-based approaches. Machine learning models analyze vast amounts of unstructured text, while lexicon-based methods use predefined sentiment dictionaries to identify sentiment in the text (S. Khan et al., 2018).

Summarization techniques, divided into extractive and abstractive methods, condense large volumes of text into concise summaries. Extractive methods select existing sentences or phrases directly from the original text, while abstractive methods generate new sentences that capture the essence of the content in a more readable form (Surbhi Bhatia et al., 2020).

Graph-based and Principal Component Analysis (PCA)-based techniques have been applied to both summarization methods to improve the quality and relevance of summaries (Surbhi Bhatia et al., 2020). Graph-based methods analyze relationships between sentences or words, while PCA-based methods focus on identifying main topics or components in the text.

Applications of opinion mining and summarization are widespread and critical across industries. In business, these techniques analyze customer reviews, feedback, and social media posts, providing insights into product performance, customer satisfaction, and emerging trends. They also contribute to recommendation systems, suggesting products or services based on sentiment analysis and summarized opinions (Vijay B. Raut, 2014). In natural language processing and text mining, these techniques efficiently process large amounts of unstructured text data and extract actionable insights (S. Khan et al., 2018). By automating sentiment analysis and text summarization, businesses can save time, resources, and improve decision-making processes.

2.12. Challenges and Limitations in text analytics:

Although text analytics has the potential to extract valuable insights from unstructured data, especially in industrial applications, it encounters several challenges that hinder its effectiveness. One such challenge is predictive analytics, which uses historical text data to forecast future trends but often suffers from the inherent variability and noise in unstructured text data. Text normalization, the process of converting raw, inconsistent data into a standardized format, is another complex task when dealing with diverse sources like customer reviews, emails, or social media posts. Without effective normalization, the analysis becomes less accurate and impedes the extraction of precise insights (Rajshekhar et al., 2016).

Another issue lies in semantic search capabilities, crucial for understanding the meaning behind the words in unstructured text. Traditional search engines primarily focus on keyword matching, whereas semantic search aims to capture context and intent, offering more meaningful results. Implementing robust semantic search techniques in large-scale text analytics systems remains a significant challenge. This complexity is further amplified by the need to transform unstructured text into structured data using sophisticated natural language processing (NLP) and machine learning (ML) techniques, which have their own limitations, including handling ambiguous language, context-dependent meaning, and scalability issues (Moreno-Sandoval & Redondo, 2016; Kalra & Agrawal, 2019).

To overcome these challenges and improve the accuracy and reliability of text analytics, researchers suggest developing interactive semantic search tools and adopting multistage approaches. These methods would enable analysts to refine search results and conduct more nuanced analysis by integrating multiple techniques such as machine learning, statistical methods, and visualization tools (Rajshekhar et al., 2016). These tools could offer more refined results in applications such as email analysis, customer feedback evaluation, and sentiment analysis, assisting businesses in making better data-driven decisions.

Despite these challenges, text analytics continues to play a vital role in enhancing business decision-making and performance. As organizations increasingly depend on unstructured data from sources like social media, reviews, and customer interactions, text analytics remains essential for understanding customer sentiments, identifying emerging trends, and making informed decisions. As advancements in machine learning, NLP, and semantic search continue, the limitations of text analytics are gradually being addressed, paving the way for more accurate and effective data analysis in industrial settings.

3. Methodology:

3.1. Introduction to Methodology:

The methodology section serves as a summary and rationale for the research's selected objectives, aims, and techniques. The "Research Onion" created by Saunders in 2019, which depicts the various phases of a research project through onion layers, will serve as the foundation for the entire methodology section. The Research Onion in Figure 1 shows each step.

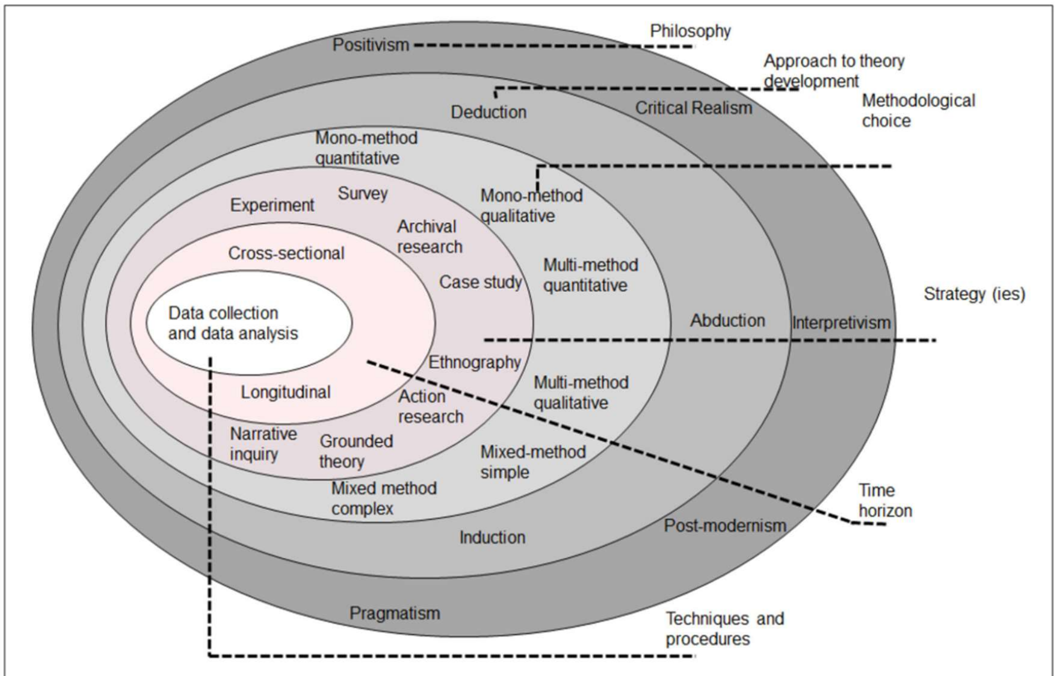

FIGURE 1 | Research onion (Saunders et al., 2019, p. 108).

Figure 1: Research Onion Source(Saunders et al., 2019)

The Research Onion (Figure 1) provides a methodical framework for handling the intricacies of study design. The six layers that make up this model each correspond to an important phase of the research process. The researcher's philosophical position, such as positivism, interpretivism, or pragmatism, which essentially influences how knowledge and reality are understood, is addressed by the Research Onion at its topmost layer (Saunders et al., 2019). These paradigms have an impact on later methodological choices, making sure that the study design fits the researcher's perspective and the nature of the investigation (Easterby-Smith, Thorpe, & Jackson, 2015). The research technique layer goes beyond this, differentiating between inductive procedures that produce theories from observed data and deductive approaches that test hypotheses (Bryman, 2016). This choice gives the study's design a rational basis and guarantees that the goals and results are consistent (Creswell & Creswell, 2018).

The inner levels concentrate on data gathering methods, strategy, and methodological decisions. While mixed and multi-method approaches incorporate both qualitative and quantitative techniques to leverage their respective strengths, mono-methods only use one of them (Tashakkori & Teddlie, 2010). Methods like surveys, experiments, and case studies are chosen according to the goals of the study and its philosophical foundations (Yin, 2014). Techniques for collecting data, such as questionnaires, observations, and interviews, are selected to complement these tactics and provide rich, pertinent data (Kumar, 2019). The Research Onion takes time horizons into account, making a distinction between longitudinal studies, which examine trends over a longer period of time, and cross-sectional research, which gather data at a single point in time. By methodically directing researchers from abstract philosophical concerns to real-world application, this tiered framework guarantees methodological rigor and offers an organized approach to research design and execution .

3.2 Research Philosophy:

The interpretivist research perspective is centered on using people's subjective experiences and interpretations to comprehend social reality. This method is especially applicable to the analysis of customer feedback since it naturally captures individual viewpoints, feelings, and experiences. Researchers can explore the complex and contextualized meanings that consumers ascribe to their encounters with companies like Scottish Power by using an interpretivist lens. This method is perfect for examining the depth of qualitative data because it recognizes that individual perceptions are influenced by a variety of intricate aspects, such as situational, social, and cultural settings (Bryman, 2016).

In contrast, optimism places a high value on analyzing quantifiable, objective facts and frequently bases its findings on data that can be measured. Though useful for seeing patterns, positivism is not as good at encapsulating the subjective nuances present in qualitative feedback. Beyond measurable measures, customer sentiments can include a range of emotional reactions and unique experiences that call for interpretation rather than quantitative research. These nuances may be missed by positivism's emphasis on objectivity, which makes interpretivism a better paradigm for this kind of study (Saunders et al., 2019).

Additionally, interpretivism stresses how meaning is context-dependent, which is important for comprehending consumer mood. Customer expectations, unique situations, and experiences with services such as those offered by Scottish Power all have an impact on feedback. Researchers can perform in-depth sentiment and topic analysis to find important themes and recurrent patterns in qualitative feedback by using interpretivist approaches. These revelations make it possible to better understand the fundamental causes of client happiness or discontent, which leads to more sensitive and efficient service enhancements (Creswell, 2018).

Furthermore, interpretivism promotes the active participation of researchers in the analytical process, enabling them to address the data in a reflexive manner and recognize their impact on interpretation. The results are richly influenced by the participants' actual experiences thanks to this cooperative relationship between the researcher and the data. Interpretivism makes it possible to engage with the data more deeply by distilling the essence of customer sentiments and viewpoints, resulting in actionable insights that accurately reflect the complexity of customer behavior (Bryman, 2016; Creswell, 2018).

3.3. Research Approach:

A deductive research methodology commences with pre-established theories or universal tenets, subsequently subjecting them to systematic assessment against empirical data to confirm or challenge the principles. This study leverages such a method to probe the interplay between customer sentiment and business results like customer loyalty and financial yield. It employs prevailing theories to shape the analysis, founded on the premise that unfavorable customer sentiment correlates with reduced levels of engagement and satisfaction. Utilizing sentiment analysis on textual feedback, the study furnishes a structured process to quantify and interpret these associations (Saunders et al., 2019). This strategy ensures the research is anchored in recognized concepts while facilitating evaluation in a practical context.

To illustrate, an analysis of customer feedback helps to understand the connection between expressed sentiment in comments and behaviors like engagement and satisfaction levels. Negative sentiments are studied for signs of dissatisfaction or decreased interaction with the business. Insights from this process can inform improvements in marketing strategies and customer experience. By adopting a deductive framework, the study systematically connects theoretical assumptions with practical insights, creating a thorough and organized method for comprehending the impact of customer sentiment (Bell et al., 2018; Robson & McCartan, 2016). This structured and clear approach makes it a valuable tool for generating insights that are both theoretically grounded and practically implementable.

3.4. Research Strategy:

The research strategy for this study applies archival research, a fitting approach for discovering patterns, themes, and insights from extant secondary data. Archival research entails systematic scrutiny of pre-collected data, like customer reviews on Trustpilot or social media posts. This method is both time- and cost-effective, as it circumvents primary data collection procedures requiring substantial planning, execution, and resource allocation (Blumberg et al., 2014). Capitalizing on an abundance of available qualitative data, researchers can unearth genuine customer stories, attitudes, and actions. Trustpilot reviews, for instance, frequently capture unedited, candid feedback from customers, rendering them a dependable resource for comprehending real-life experiences.

A key benefit of leveraging archival data lies in its capacity to sidestep certain prejudices and hurdles associated with primary data collection techniques. In contrast to interviews or questionnaires, which can be impacted by interviewer predispositions, response biases, or the artificial nature of regulated settings, secondary data echoes unplanned and organic customer expressions (Hair et al., 2015). Internet reviews and social media posts, specifically, are frequently shared instantaneously, establishing them as abundant sources of current information. This naturalness ascertains that the data accurately represents genuine customer experiences without being affected by researcher interference. Moreover, archival research promotes the scrutiny of vast data quantities over time, aiding the detection of trends and patterns that might remain undetectable in smaller, primary datasets.

It's essential to acknowledge the ethical and methodological rigor required in archival research. Ethical concerns like data privacy and informed consent need to be addressed, particularly when analyzing social media content (Blumberg et al., 2014). Researchers should also be aware of potential biases in the data, as reviews and posts might reflect experiences of highly satisfied or dissatisfied customers, which could skew results. Notwithstanding these considerations, archival research is a powerful approach for studies seeking to derive insights effectively from large datasets. This strategy supports strong qualitative research, conserves resources, and offers valuable real-world perspectives by utilizing pre-existing data.

3.5. Methodological Choice:

This study uses a qualitative mono-method approach, focusing only on textual data from Trustpilot reviews. This decision enables a thorough examination of unstructured material and is consistent with the study's objective of identifying patterns, themes, and attitudes present in consumer feedback (Denzin & Lincoln, 2018). The research maintains a clear and cohesive analytical framework by concentrating on a single approach, which is crucial for capturing the intricacies of qualitative data. Trustpilot reviews are particularly useful because they show real, uninvited user experiences. Researchers can examine not only what customers say but also how they describe their experiences thanks to the rich textual data, which provides a greater understanding of their viewpoints (Bryman, 2016).

Text mining programs like NVivo and Voyant are used to guarantee methodical and exacting examination. By reducing researcher bias and facilitating the effective processing and structuring of massive amounts of textual data, these techniques make it possible to identify recurrent themes and patterns (Guest et al., 2012). The nuances of client sentiment that quantitative methods may overlook or oversimplify are best captured by the qualitative approach. This approach is a potent choice for deciphering complicated human experiences through the words of customers since it emphasizes depth over breadth, enabling the study to unveil the subtle dynamics of customer attitudes.

3.6. Research Time Horizon:

This study focuses on consumer feedback from a particular time period utilizing a cross-sectional time horizon. A cross-sectional technique is perfect for assessing the immediate results of marketing and service strategies since it offers a snapshot of client attitudes and views at a specific moment in time (Flick, 2015). By limiting the investigation to a certain time period, the study gathers timely and pertinent data, providing actionable insights for companies wishing to make focused changes.

Cross-sectional studies are a popular choice in business research due to their proficiency in identifying patterns and trends without the need for extended monitoring or follow-up (Easterby-Smith et al., 2021). This technique proves particularly useful when the objective is to comprehend prevailing customer perspectives and actions, ensuring the outcomes remain pertinent to ongoing promotional initiatives. Utilizing this strategy, the study concentrates on providing insights that are pragmatic and firmly rooted in the current market context (Saunders et al., 2019).

3.7. Data Collection and Data Analysis:

In this study, data is amassed employing Octoparse, a web scraping instrument proficient in extracting customer reviews from Trustpilot. This tool enables the methodical compilation of a substantial data volume, guaranteeing thorough coverage of customer feedback across diverse dimensions. Utilizing web scraping, the study encompasses an extensive spectrum of views and encounters, indispensable for crafting an all-encompassing analysis (Blumberg et al., 2014). After data compilation, a purification procedure follows, eliminating inconsistencies and immaterial content to warrant the dataset's precision and applicability (Creswell & Creswell, 2018). This preprocessing phase is critical in readying the data for valuable analysis and diminishing noise that might contort the results.

NVivo is used for data analysis, conducting thematic analysis to categorize and pinpoint recurring themes within customer feedback, like satisfaction with service quality or responsiveness (Miles et al., 2014). Voyant is employed for word frequency analysis, creating visualizations such as word clouds that emphasize frequently mentioned terms, providing an intuitive summary of customer concerns and preferences (Kairaitytė-Užupė et al., 2023). Sentiment analysis tools further refine the analysis by classifying feedback into positive, neutral, or negative categories, enabling quantification of customer sentiment and identification of areas for improvement. Together, these techniques ensure a comprehensive understanding of customer feedback, delivering actionable insights for businesses.

4. Analysis and Findings:

This chapter provides for a concise and centered demonstration of analysis by presenting the results and discussion in the order of the research questions, aims, and objectives.

4.1. Word Frequency:

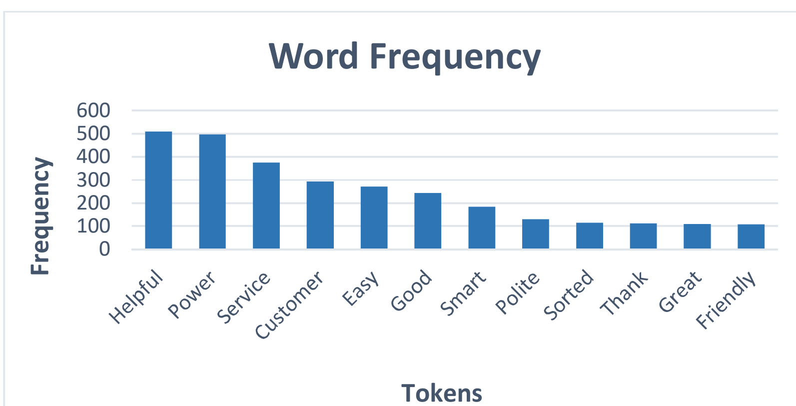

Figure 2: Word Frequency

The "Word Frequency" bar chart offers a detailed perspective on the most frequently used tokens in customer reviews, helping to identify primary themes and sentiments expressed by consumers. By examining the x-axis that features tokens like "Helpful," "Power," "Service," and "Customer," and their corresponding frequency on the y-axis, we observe a focus on customer service and satisfaction-related terms. The term "Helpful" stands out as the most frequently mentioned token, indicating that customers highly value helpful interactions with the service, making it a key factor in positive feedback.

Words like "Service" and "Customer" demonstrate that customers frequently discuss service quality and the provider's dedication to fulfilling their needs. The presence of adjectives such as "Polite," "Great," and "Friendly" underscores the significance of maintaining a positive and friendly attitude in customer support. These words illustrate that customers appreciate friendly, accommodating service, which enhances their overall experience.

Positive terms such as "Helpful" and "Thank" reinforce the sentiment that customers appreciate the support received. Service-related terms like "Sorted" and "Smart" highlight that customers prioritize effective and efficient problem-solving, leading to favorable impressions of the business. These terms suggest that customers value not only helpful assistance but also a smooth and hassle-free experience.

In conclusion, the chart emphasizes key attributes that influence customer satisfaction, including helpfulness, politeness, efficiency, and effective communication. It reveals that customers value supportive, professional, and friendly service environments, which in turn fosters their loyalty. This visualization helps businesses pinpoint which aspects of their service resonate most with customers, enabling them to focus on these areas and ensure continued satisfaction. By providing a snapshot of customer sentiments, the word frequency chart serves as a valuable tool for businesses to align their practices with customer expectations and improve service delivery.

4.2. Word Cloud:

Figure 3: Word Cloud using Voyant

In Figure 3, the word cloud aligns with the findings of more comprehensive sentiment analysis by graphically illustrating the key themes and sentiments conveyed in customer feedback. Important words like "helpful," "service," "power," "meter," "easy," and "customer" are clearly visible, indicating how frequently and how important they are in the comments. These phrases reinforce solid client connections by highlighting the need to offer dependable and easily accessible service.

Smaller yet informative words such as "problem," "account," "phone," "team," and "polite" deliver supplementary comprehension of hurdles encountered and the necessity of transparent dialogue and collaboration. They further underline the significance of preserving respect and compassion during exchanges, indicating a comprehensive perspective of both positive encounters and avenues for development.

The word cloud highlights the main ideas of providing outstanding customer service and being upbeat when interacting with customers. Phrases like "helpful," "great," and "smart" convey a dedication to fixing problems and raising client pleasure. This graphic depiction highlights the importance of constant, sympathetic, and solution-driven help while capturing the intricacy and objectives of the customer service industry.

The majority of the reviews were favorable. Words like "answered," "sorted," "email," and "spoken" complement the phrase "problem," which is used frequently, and show that customers actively contacted the business and that their issues were successfully resolved. Overall, the word cloud indicates that consumers were pleased and grateful for the service, as evidenced by the positive terms "thanks," "helpful," and "polite."

4.3. Word Link:

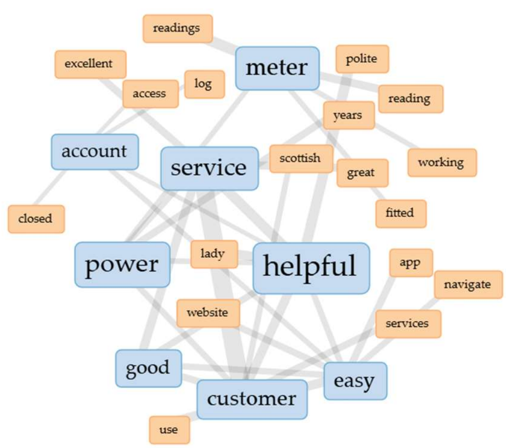

Figure 4: Word Link using Voyant

The word link visualization highlights key themes that influence the customer experience by showcasing links between different customer service components based on feedback. The term "Helpful," which represents the cooperative and encouraging aspect of good customer service, is at the center of the network. This node links to words like "easy," "service," and "great," emphasizing the value of prompt assistance, user-friendly interface design, and effective customer service. When combined, these relationships show a service environment that places an emphasis on customer pleasure and problem-solving.

Another significant subject is "Power," which emphasizes the dependability and effectiveness of the service under discussion. Related to words like "service," "account," and "meter," it highlights the significance of precise meter readings, smooth utility administration, and effective account handling. The aforementioned links indicate a focus on providing consumers with reliable infrastructure and resources to empower them and ensure seamless and stress-free service interactions.

The third important element is "Customer," which speaks to a customer-focused method of providing services. When this node is linked to terms like "good," "easy," and "website," it suggests a dedication to digital convenience, accessibility, and simplicity. These links emphasize how crucial it is to provide services that are user-friendly and customized to match client needs in order to promote pleased customers and long-term outcomes.

Peripheral terms such as "navigate," "app," and "great" enrich the network, portraying the wide-ranging facets of contemporary customer service framework. These words emphasize the significance of technology, ease of use, and professionalism, each adding to the core objective of fostering an appealing, efficient, and customer-centric setting. The visualization effectively mirrors the interconnectedness of these elements, depicting an organization committed to collaboration, dependability, and outstanding service.

According to these connections, the majority of the problems that the clients encountered were resolved promptly, and they thought the service personnel were kind and accommodating. Given that the word "app" is linked to terms like "easy," etc., the company's app appears to function effectively as well. Therefore, it is predicted that customers will be happy with Scottish Power's services.

4.4. Trend: Relative frequency:

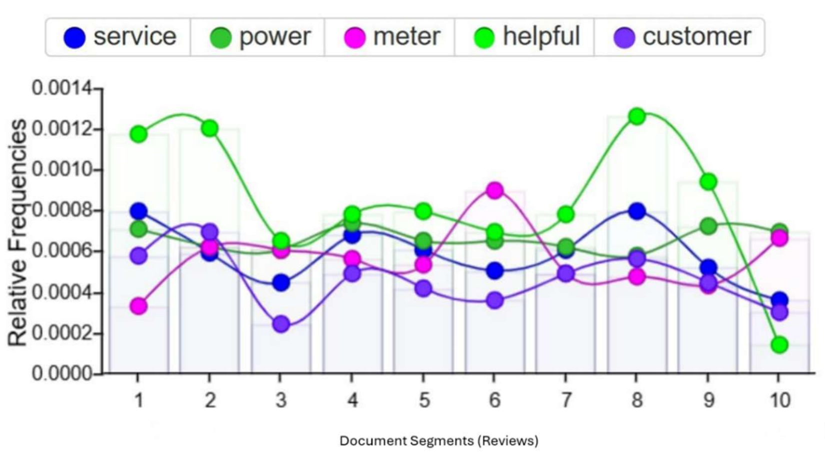

Figure 5: Trend Using Voyant

The line plot illustrates the comparative frequencies of five expressions—"service," "power," "meter," "helpful," and "customer"—analyzed across ten divisions of a document, potentially signifying customer reviews or textual input. The varying frequencies of these expressions emphasize their shifting significance within the context of the text.

The term "Customer" (purple line) exhibits a striking spike in segment 9, underscoring a substantial concentration on customer-focused subjects or matters within that text portion. This apex implies a critical discourse or insight concerning customer-associated themes in the respective segment. On the other hand, "Service" (blue line) upholds a relatively steady frequency throughout all segments, revealing its unwavering significance across the document, matching a persistent emphasis on service caliber or execution.

The "helpful" (green line) variable shows sharp peaks in segments 3 and 10, indicating focused conversations about helpfulness that may be connected to favorable feedback or particular interactions. In the meantime, "Power" (light green line) shows a stable but less prominent role in the text, remaining moderately constant over parts without significant spikes. Last but not least, "Meter" (pink line) shows sporadic spikes but maintains a generally lower frequency, suggesting its use is restricted to particular conversations. The changes in frequency between the parts may be a reflection of the evaluations' changing themes, with some portions focusing on satisfaction ("helpful") and others addressing issues like "power" or "meter." Scottish Power can examine these variances to determine shifting client preferences and enhance offerings.

Overall, the distribution of the major themes across the work is reflected in the graph. Segment 9's peak of "Customer" indicates a central conversation about customer issues, while "Service"'s constancy emphasizes its overall importance. Peak values for "Helpful" denote areas of favorable encounters, while the intermittent appearance of "Meter" points to specialized subjects. This analysis sheds light on the main ideas of the text and how significant each is in relation to the others.

4.5. Sentiment Analysis:

Sentiment analysis aims to identify and scrutinize the emotions found within textual information. This method is applied to diverse situations, including surveys, product evaluations, and customer input, to measure public sentiment. Additionally, sentiment analysis plays a pivotal part in critical responsibilities such as monitoring public opinion, social media scrutiny, and enhancing customer gratification.

Row Labels	Count of Score
Negative	591
Neutral	164
Positive	1342
Grand Total	2097

Table 1: Sentiment Count

The table 1, classifies sentiments into three main categories: Positive, Neutral, and Negative, demonstrating a variety of emotional reactions found within the data. Negative sentiments, comprising 591 instances, signify dissatisfaction, disapproval, or issues, pinpointing crucial areas of distress or unhappiness. Neutral sentiments, accounting for 164 instances, denote a noncommittal stance, lack of strong sentiment, or ambiguity. These neutral reactions constitute a relatively smaller part of the dataset, emphasizing cases where perspectives are either balanced or uncertain.

Positive sentiments, the most prevalent category with 1,342 instances, showcase extensive satisfaction, endorsement, or propitious outlooks. The substantial ratio suggests that the majority of responses convey favorable emotions, significantly surpassing both neutral and negative sentiments. The dataset, encompassing 2,097 instances, delivers a broad view of assorted perspectives, where positive emotions reign supreme, succeeded by negative sentiments, while neutral reactions constitute the least portion. These observations can support the identification of satisfaction patterns and sectors necessitating development.

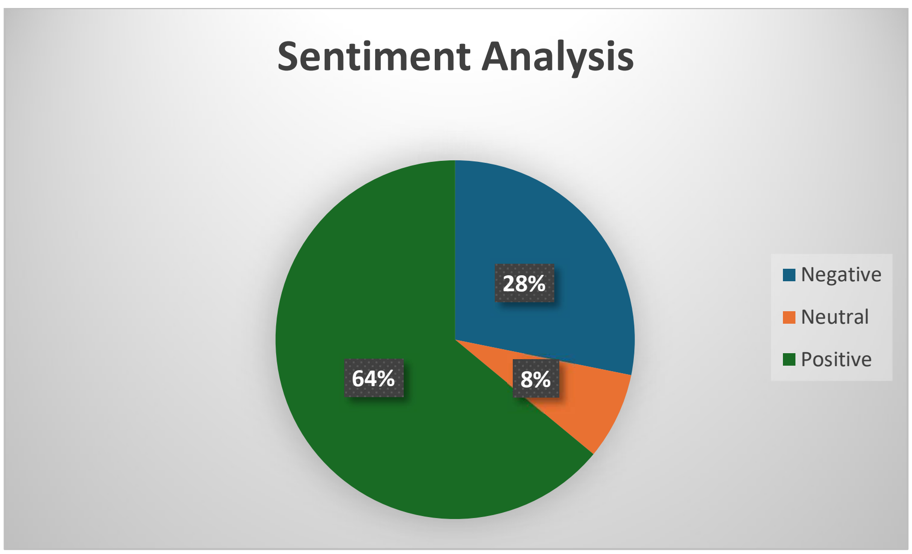

Figure 6: Sentiment Analysis

The pie chart (Figure 6) effectively conveys the spread of customer sentiment in reviews for Scottish Power, with 64% (1,342 reviews) reflecting positive sentiment. This finding aligns with previous analyses, like word cloud and trend analysis, indicating a generally content customer base. Positive sentiment is crucial for nurturing customer loyalty and satisfaction.

Although not the majority, 28% (591 reviews) of negative sentiments shouldn't be overlooked. By understanding the reasons behind dissatisfaction and actively resolving issues, businesses can reduce customer churn and convert negative experiences into improvement opportunities.

The 8% (164 reviews) neutral sentiment category presents an opportunity for businesses to refine their offerings and identify areas that may be perceived as unimpressive.

By consistently monitoring sentiment trends, companies can spot changes in customer perception. Analyzing sentiment shifts in relation to service changes, marketing efforts, or product updates can provide valuable insights. Segmenting sentiment by demographics or location may reveal areas where service delivery can be enhanced.

Overall, while the positive feedback reflects well on the company's service, the presence of negative and neutral sentiments highlights areas for improvement. Businesses should leverage these insights to reinforce successful strategies while addressing potential issues to foster a more satisfying customer relationship.

4.6. Thematic Analysis:

A thematic analysis was conducted on both positive and negative reviews of Scottish Power to identify the key themes emerging from customer feedback. Positive reviews often highlighted efficient service, reliability, and helpful customer support, while negative reviews focused on billing issues, delays in response, and communication challenges. This analysis provided a comprehensive understanding of customer sentiments and areas for improvement.

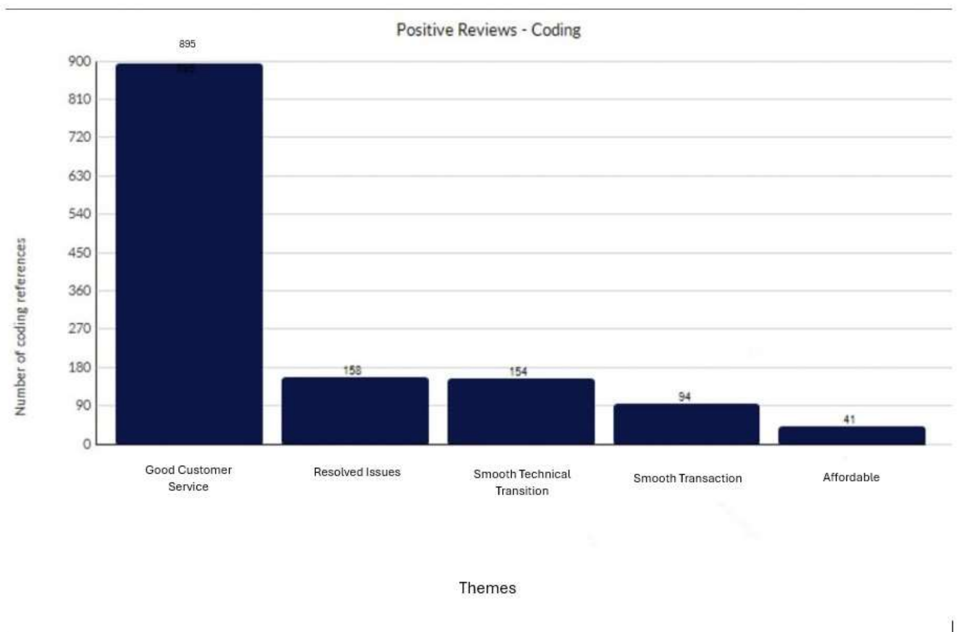

Figure 7: Thematic Analysis (Positive Reviews)

The chart (Figure 7) offers a thorough analysis of the topics found in Scottish Power's favorable ratings. With 895 references, the dominant theme, "Good Customer Service," was the most often mentioned factor contributing to customer satisfaction. This suggests that consumers value Scottish Power's customer care agents' professionalism, consideration, and helpfulness very much. Positive ratings and client loyalty are undoubtedly influenced by excellent customer service.

"Resolved Issues," the second major theme, was brought up 158 times, emphasizing that when consumers' issues or grievances are effectively resolved, they are satisfied. This indicates that prompt and efficient resolution of client complaints greatly enhances positive feedback.

Customers enjoy smooth processes, whether they are associated with technical changes, invoicing, or service activation, according to the terms "Smooth Technical Transition," which has been quoted 154 times, and "Smooth Transaction," which has been mentioned 94 times. These themes emphasize how crucial hassle-free interactions and effective operations are to preserving client happiness.

Lastly, the subject of "Affordability," despite being stated 41 times less frequently, contributes to favorable opinions. Consumers are more likely to be satisfied if they believe Scottish Power's services are reasonably priced.

Overall, the theme analysis shows that Scottish Power's outstanding service quality, efficient problem solving, seamless operations, and affordable prices are what attract customers. Scottish Power can maintain client loyalty and happiness by keeping up its strong performance in these areas.

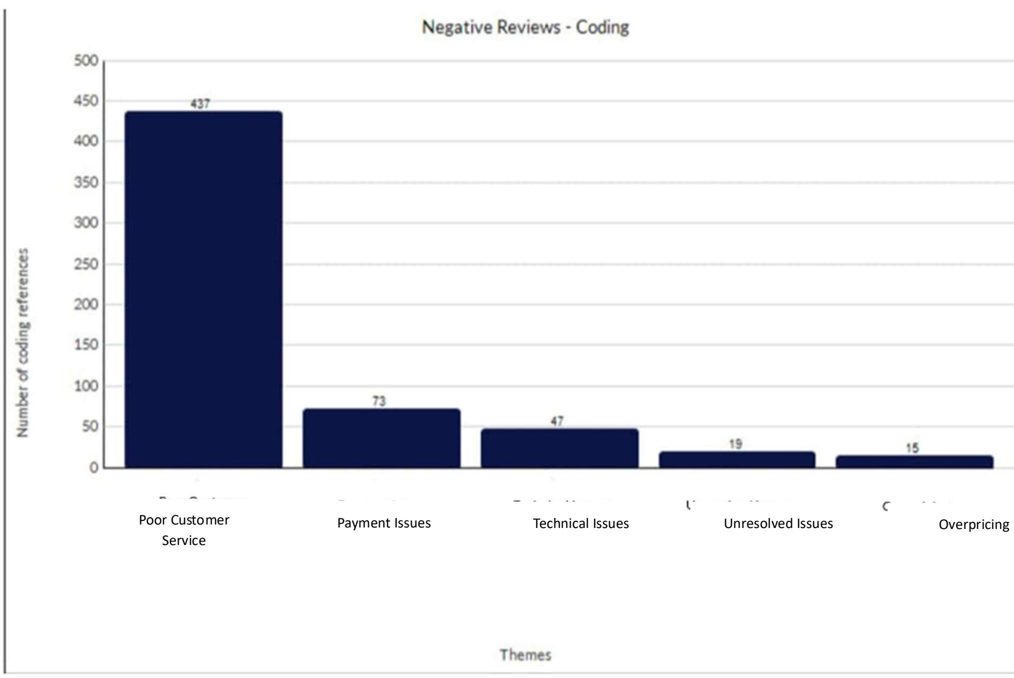

Figure 8: Thematic Analysis (Negative Reviews)

The "Negative Reviews - Coding" graphic (Figure 8) offers a thorough analysis of the themes found in customer complaints. Poor customer service, payment problems, technical problems, unresolved problems, and overpricing are some of these themes. With an astounding 437 code references, "Poor Customer Service" is the most important issue among them. This suggests that consumers are generally unhappy with the level of service they received. With 73 citations, "Payment Issues" is the second most cited theme after this, emphasizing issues with the payment procedure. While "Unresolved Issues" and "Overpricing" are less common but still exist with 19 and 15 references, respectively, "Technical Issues" is another noteworthy issue with 47 occurrences.

The most prevalent theme, "poor customer service," highlights a serious weakness in the way the business engages with and assists its clients. Consumers seem to be disappointed with the quality of service or feel ignored. This problem is so much more important than any other theme that it ought to be the main focus of development efforts. Second, payment problems can be a sign of issues with transactions, refunds, or bills, which irritates customers even more. Technical concerns, such as usability issues or system malfunctions, also contribute to the discontent.

Overpricing and Unresolved Issues are less common but equally important issues. Unresolved complaints raise the possibility of improper follow-up and resolution procedures, which may leave a bad image that lasts. Despite being the least reported, overpricing raises questions about perceived value for money and, if left unchecked, may turn off potential clients.

The most prompt issue, inadequate customer service, requires urgent attention in light of these findings. Prioritizing training, improved support systems, and faster response times will help improve the quality of customer service. In order to find and fix inefficiencies, investigations into technical and payment problems must also be conducted. Lastly, procedures should be put in place to guarantee prompt handling of grievances and pricing transparency in order to foster confidence and stop more unfavorable reviews. By addressing these problems, the business may keep its clientele and greatly increase customer satisfaction.

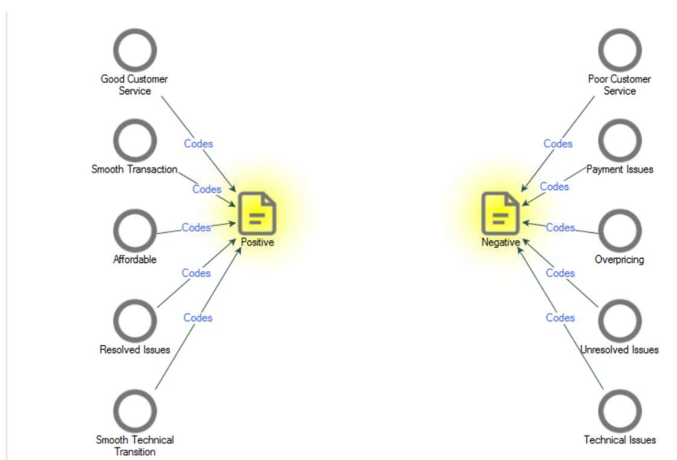

Figure 9: Node Graph (Thematic Analysis)

The nodes graph (Figure 9) shows a well-structured depiction of customer feedback, with two center yellow nodes representing the two primary sentiments: positive and negative. Within each sentiment, these nodes are further linked to particular themes (gray nodes) that emphasize the primary feedback regions. The hierarchical layout clearly visualizes consumer experiences, with themes extending from the primary sentiment nodes.

The positive sentiment node records comments from clients expressing contentment or positive experiences. Good customer service, a seamless transaction, affordability, resolved issues, and a smooth technical transition are the five main elements that it is associated with. These themes highlight the company's strong points. For example, "Good Customer Service" shows that consumers value the caliber of assistance teams' interactions. "Smooth Transaction" emphasizes easy and effective purchasing procedures, but "Affordable" alludes to favorable opinions about prices. Furthermore, "Resolved Issues" indicates that clients are satisfied when their issues are successfully resolved, and "Smooth Technical Transition" denotes a smooth system change or technical implementation experience.

The negative sentiment node, on the other hand, shows comments from clients who are unhappy or frustrated. It is related to topics like unresolved issues, overpricing, poor customer service, payment problems, and technical problems. Each of these themes highlights particular issues. "Poor Customer Service" refers to concerns over ineffective or inefficient assistance, whereas "Payment Issues" covers issues concerning invoicing, reimbursements, or other financial dealings. "Overpricing" draws attention to worries about allegedly exorbitant prices, and "Unresolved Issues" expresses discontent over issues that have not received enough attention. Finally, "Technical Issues" refers to problems or unsuccessful attempts to use technological systems or services.

This graph is a helpful tool for classifying and evaluating consumer reviews. It enables companies to pinpoint their areas of strength (emphasized by positive feedback) and areas in need of development (emphasized by negative feedback). The customer experience can be greatly improved by resolving negative themes like unsolved issues or technical challenges, while addressing positive themes like affordability and seamless transactions can strengthen customer loyalty. Through the systematic visualization of feedback, firms may monitor progress, prioritize actions, and strive to improve the customer journey.

4.7. Impact on Client and Profit Margin:

1. Analysis of the Impact of Online Reviews on Consumer Behavior:

Online reviews have been shown time and again to have a big impact on consumers' purchasing decisions. Online reviews of other people's experiences are frequently used by consumers to help them make well-informed judgments regarding the goods and services they select. Both favorable and negative reviews are important in this situation, although the latter usually have a greater effect on customer loyalty.

2. The Influence of Negative Reviews on Client Patronage:

Negative feedback can dissuade prospective clientele from associating with a business, regardless of the preponderance of positive reviews. Negative reviews frequently underscore particular problems encountered by consumers, making potential customers more vigilant. In Scottish Power's case, negative reviews might deter existing customers from maintaining allegiance and impede the onboarding of new clients. Consequently, while the cumulative influence of affirmative reviews may draw in more patrons, the persistent negative reviews could considerably counterbalance this advantage.

3. Positive Reviews and Their Role in Maintaining Patronage:

Scottish Power's preponderance of positive reviews is advantageous. The shared favorable experiences among patrons foster customer retention and draw in new customers. This feedback emphasizes the organization's strong points, bolstering its reputation in the marketplace. Hence, the surge in new clientele and preservation of present ones will likely culminate in enhanced profit margins for the company.

4. Profit Margin Dynamics:

Negative reviews have a detrimental impact on customer patronage, even though favorable evaluations are more common. Because negative evaluations have a stronger psychological impact on customer behavior, they may delay the growth of the company's profit margin. Profit margins may continue to increase, albeit at a slower rate, according to this phenomena, where the larger impact of negative evaluations partially offsets favorable influences.

4.8. Business Model:

A business model is crucial in describing how an organization generates, provides, and captures value. Scottish Power can benefit from utilizing the EFQM (European Foundation for Quality Management) model as a strategic tool to enhance customer satisfaction and overall profitability. This model offers a comprehensive framework emphasizing continuous improvement and a customer-centric approach, which are vital for boosting business performance and delivering value to stakeholders (Spencer, 2013).

The EFQM model comprises nine core criteria divided into two main categories: enablers and results. Enablers include leadership, strategy, people, partnerships & resources, and processes, which establish a robust foundation for organizational success. These factors aid businesses in executing sustainable strategies and preserving operational excellence (Asif, 2015). The results criteria emphasize customer outcomes, people outcomes, societal impact, and business outcomes, with the customer results criterion carrying the highest score weight. This accentuates the significance of customer satisfaction, loyalty, and overall experience in achieving long-term success (Hides et al., 2004).

For Scottish Power, the EFQM model can reveal areas where quality management may be deficient and help pinpoint performance gaps. By aligning operations and services with customer expectations, the company can more effectively tackle areas needing improvement. This alignment can lead to augmented customer value, higher satisfaction, and stronger brand loyalty. Implementing the EFQM model enables Scottish Power to embrace a systematic approach to continuous improvement, ensuring that customers' needs and expectations are met consistently, thereby increasing the organization's efficiency and profitability (Porter & Tanner, 2012).

Moreover, the customer-centric approach inherent in the EFQM model encourages organizations to engage with customers, gather feedback, and utilize it to refine and enhance their offerings. By prioritizing improvements in customer outcomes, Scottish Power can deliver superior services, reduce customer churn, and gain a competitive advantage in the energy sector. Regularly evaluating and refining processes will help the company maintain high performance standards, respond to evolving customer needs, and stay ahead of market trends, ultimately fostering sustained business success and increased profitability.

4.9. Market Strategy for Improvement

A clearly defined market strategy is vital for converting the business model into actionable steps that produce measurable results. Scottish Power should create a comprehensive marketing strategy that includes key elements such as brand objectives, target customer segmentation, marketing channels, and performance metrics.

1. Target Audience and Brand Positioning:

Scottish Power should concentrate on establishing itself as a customer-focused energy supplier that places a high value on sustainability and dependability (Prahalad & Ramaswamy, 2004). Segmentation should target environmentally conscious consumers and small businesses prioritizing affordability and reliability, while customer-focused initiatives like addressing feedback and ensuring transparent communication will build brand trust (Rust et al., 2000).

2. Marketing Channels:

Leveraging a blend of digital and traditional marketing channels can boost overall visibility and engagement (Chaffey & Ellis-Chadwick, 2019). Advertisements on television and in print can continue to reach audiences who are less tech-savvy, while social media, SEO, and email campaigns can target tech-savvy consumers (Ryan, 2014). Further enhancing reputation can be achieved by interacting with clients on review sites and answering their comments (Solomon, 2015).

3. Key Performance Indicators (KPIs):

Scottish Power should measure success through KPIs like customer acquisition and retention rates, net promoter scores (NPS), and sentiment analysis of online reviews, offering insights into market strategy effectiveness and areas for improvement (Reichheld, 2003).

Scottish Power can establish a reputation for quality, boost client loyalty, and achieve steady profit margin growth by combining the EFQM model with a strong marketing plan.

4.10. Results and Discussion:

The findings highlight the substantial effect of online reviews on consumer behavior, with negative reviews having more significant consequences for Scottish Power's profitability and reputation. While positive reviews contribute to customer retention, addressing negative feedback is critical for the company's growth and customer satisfaction.

Implementing the EFQM model can help Scottish Power enhance customer satisfaction and quality management, identifying areas for continuous improvement and aligning operations with customer expectations. Since the EFQM model prioritizes customer results, addressing customer concerns becomes vital in preventing negative reviews.

A customer-centric marketing strategy can also improve brand loyalty and reputation for Scottish Power. This plan should include transparent pricing, clear communication, and customer-focused activities to build trust and credibility. Enhancing engagement through personalized interactions and marketing initiatives can strengthen consumer relationships and retention.

To tackle negative reviews proactively, Scottish Power can employ predictive analytics to anticipate and address customer concerns before they escalate. Efficient feedback systems and responsive customer service programs can resolve issues promptly and reduce negative reviews.

Combining customer feedback with advanced analytics tools will allow Scottish Power to adapt its strategies continuously, resulting in improved customer loyalty, satisfaction, and profitability. This approach will strengthen the company's reputation, foster strong customer relationships, and drive sustainable growth in the competitive energy market.

5. Conclusion and Recommendation:

5.1. Conclusion:

This dissertation emphasizes how important consumer feedback is in influencing business strategies and consumer decisions, especially when it comes to online reviews and social media input. The necessity for companies to utilize unstructured customer data is highlighted by the growing dependence on these platforms. Many firms, however, find it difficult to use this input effectively, exposing a disconnect between data gathering and useful insights. This disparity emphasizes the value of technologies such as text analytics, which can convert enormous datasets into insightful data for improved decision-making.

The study found that natural language processing (NLP) and text analytics were important tools for examining consumer feedback. With the use of these tools, businesses can glean actionable themes, attitudes, and patterns from unstructured data. Positive comments frequently emphasized outstanding service quality, smooth operations, and pricing, according to the data. Conversely, unfavorable opinions centered on persistent problems like subpar customer service, unresolved grievances, and alleged overcharging. Retaining client loyalty and improving organizational effectiveness require addressing these areas of discontent.

Customer behavior has been proven to be greatly influenced by online reviews, with negative evaluations having a greater effect than favorable ones. In addition to discouraging new clients, negative opinions run the risk of weakening the loyalty of current clients. This emphasizes the necessity of proactively addressing unfavorable reviews for companies such as Scottish Power in order to reduce their influence on profitability. Organizations can lessen the negative consequences of such evaluations by concentrating on addressing customer complaints and improving the caliber of interactions.

The study suggested a structured framework that incorporates text analytics insights into company decision-making in order to address these issues. This method places a strong emphasis on doable tactics including using predictive analytics to foresee client wants, improving feedback systems for in-the-moment service enhancements, and creating clear pricing structures to build confidence. The recommendations for Scottish Power in particular included resolving major areas of unhappiness found in consumer input, improving invoicing procedures, and training customer support workers.

It was also suggested that quality management frameworks, such the EFQM model, be strategically integrated to assist match consumer expectations with company goals. When coupled with a clear market strategy that places a high value on sustainability and customer involvement, this method offers companies a road map for improving service delivery and preserving competitive advantage. The EFQM model is a useful tool for businesses looking to enhance operations and promote long-term success because of its emphasis on stakeholder value and continual improvement.

In conclusion, the dissertation emphasizes how using advanced analytics to utilize client input can have a transformative effect. Using data-driven strategies and organized frameworks, companies like Scottish Power may enhance customer satisfaction, boost service quality, and attain long-term profitability. These revelations highlight the need for businesses to adjust to changing consumer demands by leveraging feedback as a spur for innovation and expansion as well as a performance indicator.

5.2. Recommendations:

1. Training of Customer Service Personnel:

Scottish Power should to spend money on thorough training courses for its customer support staff. In order to improve customer engagement, these programs ought to concentrate on fostering interpersonal skills, dispute resolution, and sympathetic communication. Better customer service encounters can increase client contentment and loyalty, which will eventually lessen the possibility of unfavorable evaluations.

2. **Transparent Pricing:**

Transparent pricing mechanisms should be implemented by Scottish Power in order to resolve issues regarding overpricing. Customers' trust can be increased by providing thorough explanations of costs, clear and simple billing forms, and proactive notice of rate adjustments. Potential disputes will be avoided, and the business's standing as a fair and transparent organization will be improved.

3. **Institute Customer Feedback Culture:**

A strong customer feedback system that actively invites customers to share their experiences should be put in place by Scottish Power. To obtain useful information, instruments like focus groups, surveys, and online review tracking should be used. Scottish Power will have a good grasp of customer mood and areas that need development if feedback is regularly analyzed.

4. **Predictive Analytics:**

Scottish Power can use client data to spot trends, foresee requirements, and customize services by utilizing predictive analytics. Customer loyalty and happiness can be raised through tailored products and communications that are based on predictive insights. This strategy also shows that the business respects its clients as distinct persons with particular requirements.

5. **Streamlined Feedback Mechanism:**

Establishing an effective feedback gathering and assessment framework can empower Scottish Power to manage customer issues instantly. Leveraging sophisticated technologies like sentiment analysis automation and customer experience tools can assist in recognizing trends and swiftly adapting services. This ongoing refinement method will transform the organization into a more reactive and customer-focused entity.

6. **Customer Care Initiative:**

A targeted customer service campaign should be started by Scottish Power to mitigate the effects of unfavorable evaluations. A specialized resolution team could be part of this program, whose job it is to promptly and completely handle serious complaints. Prioritizing the concerns brought up in unfavorable reviews will help the business increase customer satisfaction and lessen the negative effects on its earnings and reputation.

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Appendices:

1. Sentiment Analysis:
   https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B4DD48262-73D5-4B0C-B40F-02123B505673%7D&file=Sentiment%20Analysis.xlsx&action=default&mobileredirect=true

2. Word Frequency:
   https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B1C1CA72C-DEDE-47A5-8E63-75A348ACC8BA%7D&file=Word%20Frequency.xlsx&action=default&mobileredirect=true

Ethics Approval Certificate:

Coventry University

Certificate of Ethical Approval

Applicant:	Rashmi Gajbhiye
Project Title:	"Leveraging Text Analytics for Customer Insight: A Study of Market Strategy Improvement through Online Reviews and Social Media Feedback – A Case Study of Scottish Power"

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	17 Oct 2024
Project Reference Number:	P181173